

Short-Horizon Air Quality Forecasting: A Comparative Study of Regression Ensembles with Meteorological and LLM-Derived Features

Rashmi Sinha¹ Yashavi Tyagi² Parth³ Tapas Dev Yadav⁴

¹Department of Computer Science, IIIT Delhi {2025414,2025574,2025360,2025511}@iiitd.ac.in

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Abstract. Air pollution is a key environmental risk factor in Indian cities, and short-horizon forecasts of pollutant concentration are the key input for issuing timely public-health advisories. We formulate short-term ($t+1$ to $t+24$ hour) PM_{2.5} forecasting as a supervised regression and threshold-classification task, using publicly available government- or research-sourced quantitative air-quality datasets paired with meteorological covariates. We establish a persistence and linear-regression baseline for the regression view and a majority-class baseline for the AQI-bucket classification view, then compare these against regularized linear models, sequence models, and tree ensembles (Random Forest and Gradient Boosting), evaluated via RMSE, MAE, and AQI-bucketed classification metrics, along with station/season subgroup and extreme-event failure analysis. As an auxiliary experimental branch, we investigate whether LLM-derived embeddings of publicly issued air-quality advisory text add predictive value on top of the numeric pipeline. Our core project does not depend on this branch succeeding.

1 Problem Statement and Motivation

PM 2.5 “is linked to increased mortality and morbidity, with increased incidences of respiratory infections, cardiovascular diseases, and lung cancer”¹. Even CPCB issues daily AQI bulletins to guide advisories. However, if we forecast pollutant concentrations ahead of time, rather than relying solely on same-day monitoring, we can publish advisories more proactively around threshold crossings into the “poor”/“severe” AQI bands. We will treat this as a supervised time series regression problem: Given pollutant data, location and weather readings up to time t , we want to predict PM_{2.5} concentrations at horizon $t+h$, $h \in [1,24]$, and then map these forecasts onto standard AQI buckets for threshold-based decision relevance.

2 ML Task and Research Questions

Let $x_t \in \mathbb{R}^d$ be lag, rolling-statistic, and meteorological features at hour t ; we learn $f_\theta : x_{t-k:t} \mapsto \hat{y}_{t+h}$, where $x_{t-k:t}$ is the vector of all collected parameters from time $t-k$ through t , and $\hat{y}_{t+h}$ is the predicted AQI. Research questions: (1) how much do weather and lag features improve over a persistence baseline per horizon; (2) where does performance degrade – station/city, season, or extreme-pollution events (winter inversion, festival spikes); (3) (optional) does an LLM-embedding of advisory text add uplift over the numeric pipeline, or is the latter sufficient alone.

3 Related Work

Patel et al. (2025) conducted a systematic study on PM_{2.5} and PM₁₀ concentration prediction using machine learning and deep learning, showing that LSTM outperforms traditional methods. Saminathan and Malathy (2023) proposed an ensemble-based classification approach for PM_{2.5} forecasting using meteorological data, employing XGBoost and a majority-class baseline. We similarly use XGBoost and perform classification on AQI buckets, as these are standard strong baselines preceding deep sequence models; we treat them as our methodological core and sequence models/text embeddings as clearly delineated extensions.

4 Dataset and Data Understanding

We use existing public datasets rather than collecting new data: (1) the **India Air Quality Index Dataset [2023–2025]** (Udayana, 2025), with 235,000+ AQI records across Indian cities; (2)

ERA5 hourly time-series reanalysis data (Copernicus Climate Change Service, 2018), the fifth-generation ECMWF global climate/weather reanalysis; (3) (optional, for reference) **15-minutely CPCB air-quality data** (Vivek, 2026). The AQI dataset will be mapped to ERA5 using station coordinates and timestamps, preferred for direct relevance to Indian public health. Planned EDA: distributions and sampling frequency; imputation on missing-value patterns; correlation and leakage checks; diurnal and seasonal periodicity checks.

5 Methodology

Baselines: persistence ($\hat{y}_{t+h} = y_t$) and linear regression on lag/weather features for the regression view; majority-class baseline for the AQI-bucket classification view. **Core models:** regularized linear modelling (Ridge, Lasso), Random Forest (Breiman, 2001), and XGBoost (Chen and Guestrin, 2016) for gradient boosting; a 1D-CNN-LSTM is a stretch goal. **Auxiliary branch:** frozen pretrained sentence embeddings of public advisory text appended to the best numeric model, with source, fine-tuned status, rationale, baseline comparison, and limitations (leakage risk, granularity mismatch) documented explicitly – this is not core to the project. Features include rolling statistics and cyclical time encodings; splits are strictly chronological to prevent look-ahead leakage, with hyperparameters tuned only on the validation split.

6 Evaluation and Validation Strategy

For regression: RMSE, MAE, MSE per horizon. For AQI buckets: macro-precision, recall, and F1. Results are further stratified by season, weather, and location. The test split is touched once, at the end, after all tuning on the validation split.

7 Expected Challenges

(1) Missing or drifting data may require carefully measured and justified imputation; (2) extreme pollution events (e.g., Diwali) are the most important yet most difficult to predict, being outliers; (3) if the LLM-text branch proves infeasible, we report a scoped negative result and rely on the numeric pipeline as the core contribution.

8 Work Plan and Division of Responsibilities

Responsibilities are divided among the four team members by workstream. **Rashmi Sinha** leads data acquisition and preprocessing, including merging the AQI and ERA5 datasets, handling missing values, and engineering lag and rolling-statistic features. **Yashavi Tyagi** implements and tunes the baseline and regularized linear models (persistence, majority-class, Ridge, Lasso). **Parth** develops the tree-ensemble models (Random Forest, XGBoost) across all forecast horizons and pursues the 1D-CNN-LSTM as a stretch goal. **Tapas Dev Yadav** owns the evaluation framework (RMSE, MAE, macro-precision/recall/F1), conducts the station/season and extreme-event subgroup analysis, and leads the optional LLM-embedding auxiliary branch. All members collaborate in the final phase to consolidate results and prepare the final report.

References

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